

Conditional Probability

Lecture 3

Topics in this lecture

- Conditional probability อ. หน้าจะฟัน ยี่สิบ หนึ่งร้อยหก
- Independence of events อ. ยี่สิบ หนึ่งร้อยหก ยี่สิบ หนึ่งร้อยหก
- Sequence of independent experiments
 ↳ ลำดับของเหตุการณ์ที่ ยี่สิบ หนึ่งร้อยหก ยี่สิบ หนึ่งร้อยหก

เหตุการณ์หนึ่งเกิดขึ้นแล้ว จะส่งผลให้โอกาสเกิดของอีกเหตุการณ์หนึ่งเปลี่ยนแปลงไป

Conditional Probability

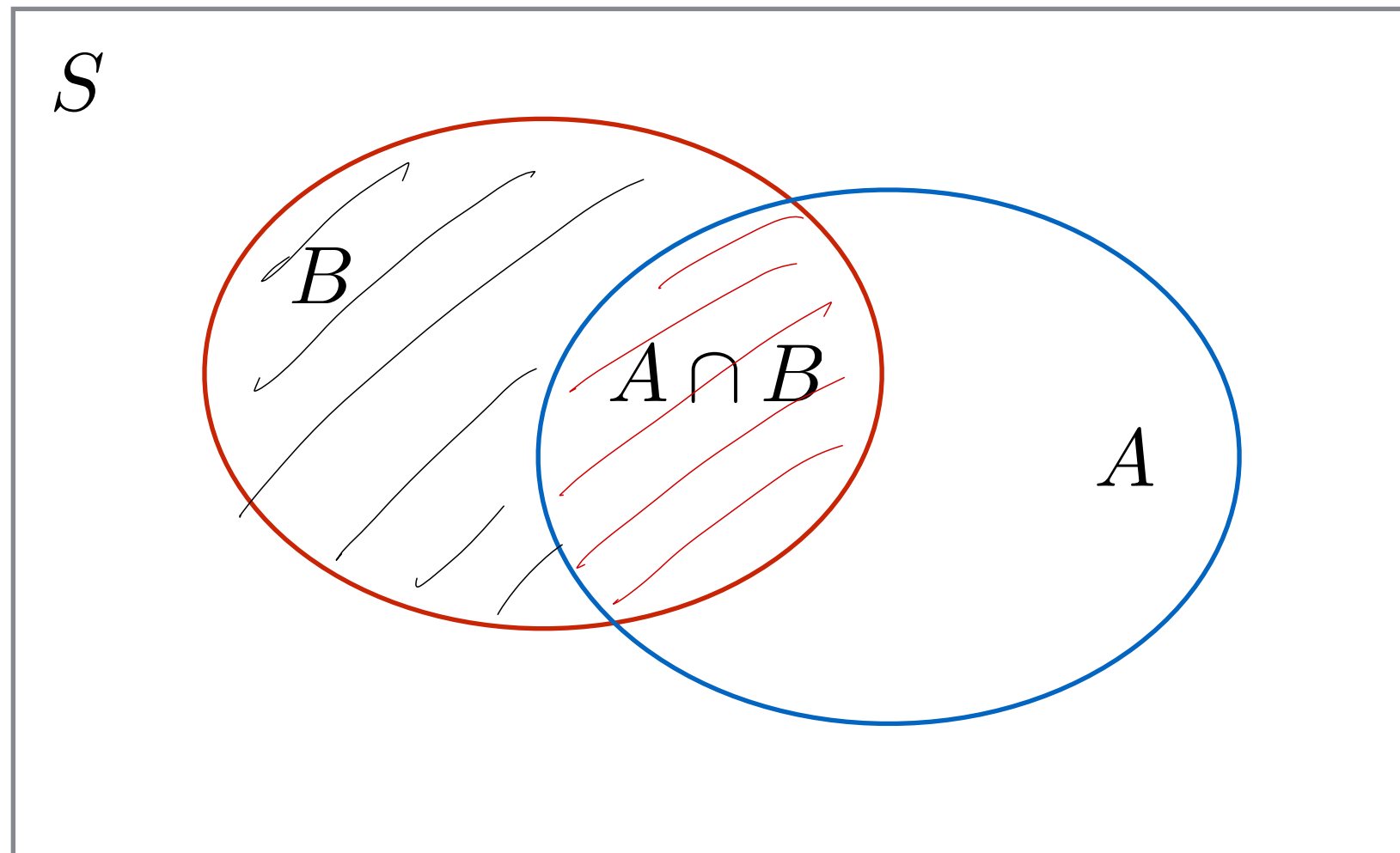
- Does knowledge about the occurrence of one event, say B, alter the likelihood of occurrence of the other, say A?
- Conditional probability is defined as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \text{ for } P(B) > 0$$

โอกาสเกิด A เมื่อ given B

Graphical Interpretation

$\tilde{P}(B) \text{ ที่ } \tilde{P}(A)$
 $= P(A|B)$
 $\tilde{P}(A) \text{ ที่ } \tilde{P}(B)$
 $= P(B|A)$
 $\hookrightarrow \frac{P(A \cap B)}{P(A)}$



$$\frac{n_{A \cap B}}{n_B} = \frac{n_{A \cap B}/n}{n_B/n} \rightarrow \frac{P(A \cap B)}{P(B)}$$

Alternative Expressions

$$P(A \cap B) = P(A|B)P(B)$$

$$P(A \cap B) = P(B|A)P(A)$$

Example 3.1

โยนเหรียญ 3 ครั้ง

หา. ถ้าจะ ได้หัว 3 ครั้ง

- Toss a coin 3 times. What is the conditional probability of getting all heads given that

โยนเหรียญ :

(a) The first toss lands on head $\rightarrow$ ครั้งแรกโยนได้หัว

$$\Omega = \{HHH, HHT, HTH, THH, TTH, THT, HTT, TTT\}$$

$$A = \{HHH\} \quad \# \text{ event}$$

$$B = \{HHH, HTH, HHT, HTT\} \quad \# \text{ โยนได้หัว } a) \rightarrow P(B) = 4/8$$

$$A \cap B = \{HHH\} \quad \rightarrow P(A \cap B) = 1/8$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{1/8}{4/8} = 1/4 \quad \#$$

Example 3.1

- Toss a coin 3 times. What is the conditional probability of getting all heads given that

$$\Omega = \{HHH, HHT, HTH, THH, TTH, THT, HTT, TTT\}$$

(b) At least 1 toss lands on head *បើយើងបាន ១ ឬ ២ ឬ ៣ ក្បាល ចុះ ៗ ក៏បាន*

$$A = \{HHH\}$$

$$C = \{HHH, HHT, HTH, THH, TTH, THT, HTT\}$$

$$A \cap C = \{HHH\} \rightarrow P(A \cap C) = 1/8$$

$$P(A|C) = \frac{P(A \cap C)}{P(C)} = \frac{1/8}{7/8} = 1/7 \neq$$

Example 3.2

- Roll a die twice and observe the sum of the marks from two rolls. Find the conditional probability that the sum of the marks is greater than 7 given that
 - (a) The outcome of the first roll is 3
 - (b) The outcome of the first roll is 6

Example 3.3

- A ball is selected from an urn containing two black balls, numbered 1 and 2, and two white balls, numbered 3 and 4. The number and color of the ball is noted. Find $P(A|B)$ and $P(A|C)$ where A, B, C are

- A = "Black ball is selected" ๔๒๒๒๒๒๒๒

- B = "Even-numbered ball is selected" ๔ ๒๒๒๒๒๒

$$\Omega = \{(1, W), (2, W), (3, b), (4, b)\}$$

$$A = \{(3, b), (4, b)\}$$

$$B = \{(2, W), (4, b)\}$$

$$A \cap B = \{(4, b)\}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{1/4}{1/2} = 1/2 \quad \#$$

Total Probability Theorem

- Let $B_1, B_2, \dots, B_n$ be *mutually exclusive* events whose union equals the sample space S . We refer to these sets as a partition of S
- The event A can be represented as

$$\begin{aligned} A &= A \cap S = A \cap (B_1 \cup B_2 \cup \dots \cup B_n) \\ &= (A \cap B_1) \cup (A \cap B_2) \cup \dots \cup (A \cap B_n) \end{aligned}$$

- The probability of event A can be written as

$$P(A) = P(A \cap B_1) + P(A \cap B_2) + \dots + P(A \cap B_n)$$

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \dots + P(A|B_n)P(B_n)$$

Example 3.4

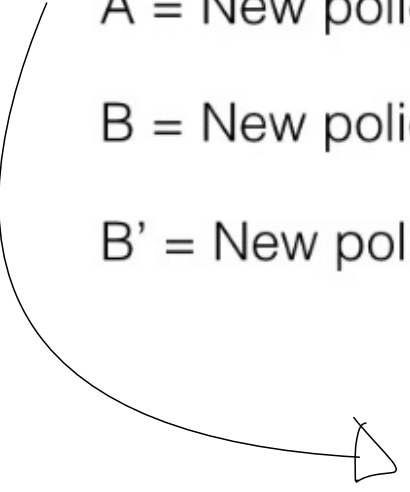
ဤ အ. ဖုန်းခံမိသူများကို ခွဲခြား၍ အန္တရာယ်ရှိသူများ နှင့် အန္တရာယ်မရှိသူများ ခွဲခြားနိုင်ပါသည်။

- An insurance company believes that people can be divided into two classes: those who are accident prone and those who are not. The company's statistics show that an accident-prone person will have an accident at some time within a fixed 1-year period with probability 0.4, whereas this probability decreases to 0.2 for a person who is not accident prone. If we assume that 30 percent of the population is accident prone, what is the probability that a new policyholder will have an accident within a year of purchasing a policy?

A = New policy holder will have accident within 1 year

B = New policy holder is accident prone

B' = New policy holder is NOT accident prone



$$P(A) = P(A|B)P(B) + P(A|B')P(B')$$

A = New policy holder will have accident within 1 year

B = New policy holder is accident prone

B' = New policy holder is NOT accident prone

$$P(A) = P(A|B)P(B) + P(A|B')P(B')$$

$$P(A|B) = 0.4 \quad \# \text{ ถ้าผู้ถือค้ายางใหม่ อยู่ในกลุ่มเสี่ยง}$$

$$P(A|B') = 0.2 \quad \# \text{ ไม่ได้อยู่ในกลุ่มเสี่ยง}$$

$$P(B) = 0.3 \quad \# \text{ ๑๕% ในกลุ่มเสี่ยง}$$

$$P(B') = 0.7 \quad \# \text{ ๗๕% ในกลุ่มไม่เสี่ยง}$$

$$P(A) = (0.4 \times 0.3) + (0.2 \times 0.7) = 0.26 \quad \#$$

Bayes' Rule

- Let $B_1, B_2, \dots, B_n$ be a partition of a sample space S . Suppose that event A occurs, the probability of event B_j given A is

of Bayes'

$$P(B_j|A) = \frac{P(A \cap B_j)}{P(A)} = \frac{P(A|B_j)P(B_j)}{\sum_{k=1}^n P(A|B_k)P(B_k)}$$

- $P(B_j)$'s are called "a priori" probabilities
- $P(B_j | A)$ are called "a posteriori" probabilities

Example 3.5

ឧទាហរណ៍ 3.4 បើ បំណង ថា បុគ្គល ម្នាក់ ដែល បាន ប្រើប្រាស់ យានយន្ត មាន ប្រសិទ្ធភាព ក្នុង ការ ប្រើប្រាស់ យានយន្ត ឬ ទេ ។

- From Example 3.4, suppose that a new policyholder has an accident within a year of purchasing a policy. What is the probability that he or she is accident prone?

ឬ ឬ $P(B|A)$

A = New policy holder will have accident within 1 year

B = New policy holder is accident prone

B' = New policy holder is NOT accident prone

$$P(A|B) = 0.4 \quad P(B) = 0.3$$

$$P(A|B') = 0.2 \quad P(B') = 0.7$$

$$P(B|A) = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B')P(B')} = \frac{0.4 \times 0.3}{(0.4 \times 0.3) + (0.2 \times 0.7)} = 0.46 \neq$$

Example 3.6

- A laboratory blood test is 95 percent effective in detecting a certain disease when it is, in fact, present. However, the test also yields a “false positive” result for 1 percent of the healthy persons tested. (That is, if a healthy person is tested, then, with probability 0.01, the test result will imply that he or she has the disease.) If 0.5 percent of the population actually has the disease, what is the probability that a person has the disease given that the test result is positive?

Example 3.7

- A ball is selected from an urn containing two black balls, numbered 1 and 2, and two white balls, numbered 3 and 4. Let the events A, B, and, C be defined as

$$\Omega = \{(1, w), (2, w), (3, b), (4, b)\} \quad \left| \quad \begin{array}{l} P(A) = 2/4 \\ P(B) = 2/4 \\ P(A \cap B) = 1/4 \end{array} \right.$$

- A = "Black ball is selected" $\rightarrow \{(3, b), (4, b)\}$

- B = "Even-numbered ball is selected" $\rightarrow \{(2, w), (4, b)\}$

$$1/4 = 1/4 \rightarrow \text{true}$$

$$P(A \cap B) = P(A)P(B) ? \quad \Rightarrow \quad 1/4 = 2/4 \times 2/4 \quad \# \text{ ไม่เป็นอิสระกัน}$$

- Are events A and B independent? $\checkmark$
- Are events A and C independent?

Example 3.7

- A ball is selected from an urn containing two black balls, numbered 1 and 2, and two white balls, numbered 3 and 4. Let the events A, B, and, C be defined as

- A = "Black ball is selected" $\{(1, b), (2, b)\}$

$$P(A) = 2/4 \quad P(A \cap C) = 2/4$$

$$P(C) = 2/4$$

- C = "number of ball is greater than 2" $\{(3, b), (4, b)\}$

- Are events A and B independent?

- Are events A and C independent? $P(A \cap C) = P(A)P(C) ? \Rightarrow$ ไม่เป็นอิสระ
 $2/4 \neq 2/4 \times 2/4$ ✗

Example 3.8

- Two numbers x and y are selected at random between 0 and 1. Let events A , B , and C be defined as follows

$$A = \{x > 0.5\}$$

$$B = \{y > 0.5\}$$

$$C = \{x > y\}$$

- Are events A and B independent?
- Are events A and C independent?

Sequence of Independent Experiments

- Suppose that a random experiment consists of performing experiments $E_1, E_2, \dots, E_n$. Let $A_1, A_2, \dots, A_n$ be events such that A_k concerns only the outcome of the k -th sub-experiment. If the sub-experiments are independent, then it is reasonable to assume that events $A_1, \dots, A_n$ are independent.
- It follows that

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1)P(A_2) \dots P(A_n)$$

Example 3.9

- Box 1 contains 5 red balls and 5 yellow balls. Box 2 contains 3 red balls and 7 white balls. Select a ball from each box. Find the probability that the 2 red balls are selected.

$$P(A) = 5/10$$

$$P(B) = 3/10$$

$$P(A \cap B) = \frac{5}{10} \times \frac{3}{10} = 0.15 \quad \#$$

Example 3.10

- Suppose that 10 numbers are selected at random from the interval $[0,1]$. Find the probability that the first 5 numbers are less than $1/4$ and the last 5 numbers are greater than $1/2$.